

Prime floaters were often marketed to investors who were used to buying more traditional floating rate notes based on LIBOR. There were periods of time when the initial coupons paid by Prime floating rate notes were in excess of what could be obtained in comparable LIBOR-based floating rate notes. This made Prime floaters relatively attractive to certain investors.

7.6 Capped Floaters

Capped floating rate notes are floating rate notes in which the coupon no longer floats when the floating rate index rises above a certain prespecified level. In this case, the coupon is fixed at the *coupon cap* for that period. The buyer of a capped floater is giving up some of his potential upside; if rates rise, the holder of the bond will never be able to earn a coupon above the coupon cap. In return for giving up this upside, the bondholder receives an enhanced spread to the floating rate index.

As an example, suppose that 3-year floating rate notes issued by a AA bank pay a coupon of 3-month LIBOR $- 15$ bps. Capped floaters issued by the same bank may pay coupons of, say, 3-month LIBOR $+ 15$ bps, capped at 7%. The terms of this note are shown in Table 7.4. The buyer of this capped floater receives an additional 30 bps margin in his coupon (i.e., the difference between coupons of 3-month LIBOR $+ 15$ bps and 3-month LIBOR $- 15$ bps) but will never be able to earn a coupon higher than 7%. By the way, what prespecified level does LIBOR have to reach in order for the coupon to no longer float?¹⁵

In general, the coupons of a capped floater are of the form LIBOR $+ y$ bps, capped at $z\%$. These coupons can be written as $\min(\text{LIBOR} + y, z\%)$. The term y is referred to as the *margin*, and the term z is referred to as the *coupon cap*. Notice that

$$\min(\text{LIBOR} + y, z\%) = \text{LIBOR} + y - \max[0, \text{LIBOR} - (z\% - y), 0]. \quad (7.1)$$

This shows that the buyer of a capped floater is actually long a floater paying a coupon of LIBOR $+ y$ and short a LIBOR cap struck at $z\% - y$. The investor receives an enhanced margin to LIBOR to compensate him for selling the LIBOR cap. This enhanced margin implies a higher initial coupon for the note holder. His risk, however, is that his coupons will be capped if rates rise high enough. In that case, he may have been better off buying a plain vanilla floater instead of a capped floater.

¹⁵The coupon will no longer float when 3-month LIBOR $+ 15$ bps is greater than 7% or when 3-month LIBOR is greater than 6.85%. If on a reset date 3-month LIBOR is greater than 6.85%, the holder of the bond will only receive a coupon of 7%.

Issuer:	AA Bank
Principal:	\$100mm
Maturity:	3 years
Coupon:	3-month LIBOR + 15 bps, capped at 7%
Resets and Payments:	Quarterly
Day Count:	Act/360
Issue Price:	100%

Table 7.4: Terms of a Capped Floating Rate Note

Capped floaters are appealing to investors who think that short-term rates will not rise as quickly as the forwards predict.¹⁶ They therefore do not mind giving away some of their upside in exchange for an enhanced margin to their coupon. Implicitly, the holder of a capped floater is also taking a view on volatility. Since the buyer of the note is selling a cap (he is giving away his upside in case LIBOR rises up too high, and he will only get coupons equal to the coupon cap), he is short volatility. That is, the value of the note will increase (decrease) when option implied volatility decreases (increases).

In the early 1990s when the structured note market was just taking off, capped floaters were very popular with investors who thought that the U.S. economy would remain in a recession. During this time the swap curve was steep, implying that the forwards were predicting that short-term rates would soon rise. Some investors thought the Fed would cut short-term rates to stimulate the economy. Some people were willing to give up some of their upside by selling caps that were high in value due to the steepness of the curve. In addition, implied cap volatility during this time was relatively high, which also made the prospect of selling caps attractive.

7.6.1 An Example: Pricing Up a Capped Floater

Recall our stylized example in which the swap curve is quoted annual bond versus 1-year LIBOR and consider the swap curve shown in Table 7.5. Suppose there is an investor who wants to purchase a capped floating rate note

¹⁶Actually, in the early days of the structured note market, it might have been more accurate to say that capped floaters were popular with investors who thought that short-term rates will not rise quickly. Some investors at this time were analyzing the pricing inherent in these notes not based on using forward curves but, rather, based on using their views on the market and the optics of the note. The fact that such notes offered a substantial pickup in the margin (e.g., 30 bps in the note in Table 7.4) was enough to induce some investors to buy the note without careful consideration of other aspects of the trade.

that matures in five years and has a coupon of $\text{LIBOR} + 10$ bps. Assume an issuer has been found who will pay $\text{LIBOR} - 15$ bps. What coupon cap must be imposed on the note so that the dealer breaks even?

In the top half of Figure 7.4, we bootstrap the curve and compute the value (from the point of view of the dealer) of paying out $\text{LIBOR} + 10$ coupons and receiving $\text{LIBOR} - 15$ coupons. Note that this is equivalent to the dealer paying out an annuity of 25 bps running. As this figure shows, these cash flows cost the dealer \$1,070,322 in net present value terms. The dealer will make up this value by buying the cap that is embedded in the structure (i.e., that the investor is shorting when he buys the capped floater). That is, the cap that the dealer purchases must be worth precisely \$1,070,322 in order for the dealer to break even. We must solve for that strike on a 5-year cap such that the cap is worth this amount.¹⁷ As shown in the bottom half of Figure 7.4, assuming the base vol curve is flat at 15%, the appropriate strike on the cap is 6.46%. Since the margin on the bond is $y = 10$ bps and the strike on the embedded cap is $z\% - y = 6.46\%$, Equation 7.1 tells us that the coupon cap is $z = 6.56\%$.

Let's now consider a second example. Suppose all the assumptions made above still hold, except that the issuer is willing to fund itself at $\text{LIBOR} - 20$ bps rather than at $\text{LIBOR} - 15$ bps. Thus, the issuer is demanding more advantageous funding for itself. In the underlying swap transaction between the dealer and the issuer, the dealer will now be receiving $\text{LIBOR} - 20$ bps instead of $\text{LIBOR} - 15$ bps. If we solve for the strike on the cap in this example, we find (in a calculation not shown here), that the strike on the embedded cap is $z\% - y = 6.23\%$. Note that the margin on the bond is still assumed to be $y = 10$ bps. Thus, from Equation 7.1 we have that the coupon cap is $z = 6.33\%$. Note that as a consequence of the issuer demanding an improved funding level for itself, the economics offered to the investor who buys the note are diminished. Instead of buying a bond that pays coupons equal to $\min(\text{LIBOR} + 10, 6.56\%)$, the investor now gets a bond that pays $\min(\text{LIBOR} + 10, 6.33\%)$. The result of improving the funding level to the issuer by 5 bps running was to lower the coupon cap by $6.56\% - 6.33\% = 23$ bps.

7.7 Inverse Floaters

Inverse floating rate notes are floating rate notes whose coupons increase (decrease) when the underlying floating index decreases (increases). Assuming

¹⁷This can be easily accomplished using a canned iteration function in a spreadsheet package.

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	4.80%	20	5.00%
2	4.88%	23	5.11%
3	5.05%	25	5.30%
4	5.19%	28	5.47%
5	5.25%	35	5.60%

Table 7.5: The Swap Curve

Time	zc	Forward	L – 15	PV(L – 15)	L + 10	PV(L + 10)
1	5.000%	5.000%	4,850,000	4,619,048	(5,100,000)	(4,857,143)
2	5.113%	5.226%	5,075,748	4,593,976	(5,325,748)	(4,820,247)
3	5.312%	5.712%	5,562,202	4,762,230	(5,812,202)	(4,976,274)
4	5.494%	6.041%	5,891,364	4,756,682	(6,141,364)	(4,958,532)
5	5.635%	6.202%	6,052,087	4,601,086	(6,302,087)	(4,791,148)
				23,333,022		(24,403,344)
				NPV (1,070,322)		
Notional		100,000,000				

Cap model									
							Forward		Caplet
Time	zc	Period	Forward	σ	d_1	d_2	premium	Caplet	in dollars
1	5.000%	0x1	5.000%	n/a	n/a	n/a	0.000%	0.000%	0
2	5.113%	1x2	5.226%	15%	-1.339	-1.489	0.031%	0.028%	28,000
3	5.312%	2x3	5.712%	15%	-0.474	-0.686	0.224%	0.192%	191,606
4	5.494%	3x4	6.041%	15%	-0.128	-0.388	0.458%	0.370%	369,690
5	5.635%	4x5	6.202%	15%	0.014	-0.286	0.633%	0.481%	481,026
Strike 6.460%			Notional		100,000,000				1,070,322
									in bps 107.0

Figure 7.4: Valuing the Cash Flows in the Capped Floater and Solving for the Strike on the Cap

the inverse floater is tied to movements in a LIBOR index, the coupon is of the generic form¹⁸

$$C\% - N \times \text{LIBOR}, \text{ subject to a minimum coupon of } 0\%. \quad (7.2)$$

In Equation 7.2, C is a fixed constant, and N is a positive integer. If $N = 1$, then the bond is referred to as an inverse floater. If N is greater than one, then the bond is referred to as a *leveraged inverse floater*. People sometimes refer to leverage inverse floaters generically as just inverse floaters.

As we have discussed previously, in an ordinary floating rate note, the reset mechanism tends to bring the value of the note back to or near par on any reset date. This is not the case with an inverse floater. From Equation 7.2, we see that as LIBOR goes up, the coupons go down, and vice versa. If LIBOR goes up high enough, the coupons will be zero. In an extremely high interest rate environment, in which all of the coupons in an inverse floater are predicted (as suggested by the forwards) to be zero, the note will trade much like a zero coupon bond, which certainly is not worth par.

As an example, consider the terms of the leveraged inverse floater shown in Table 7.6. This bond pays a coupon of $20\% - 4 \times 1\text{-year LIBOR}$, subject to a minimum coupon of zero. The first thing to note is that an option must be embedded into the structure in order to guarantee that the minimum coupon is zero regardless of what happens to 1-year LIBOR. This option (the one that guarantees that the coupon will never be negative) will be in-the-money when

$$20\% - 4 \times 1\text{-year LIBOR} < 0\%,$$

or when $1\text{-year LIBOR} > 5\%$. Thus, it is a cap that is embedded into the structure.

Note that an investor who buys the bond is actually long the cap; the cap works to the advantage of the investor to make sure that the minimum coupon level is always observed, regardless of where 1-year LIBOR sets. In addition, note that for a principal amount of \$100mm of the bond the structure requires \$400mm, not \$100mm, of a cap struck at 5%. The reason that a notional of \$400mm and not \$100mm is needed on the embedded cap is a direct consequence of the leverage factor of 4.

In order to see why \$400mm of the cap is necessary, suppose that 1-year LIBOR resets to 5.01% on a reset date. In this case, if there were no minimum coupon restriction, the coupon would reset to $20\% - 4 \times 5.01\% = -0.04\%$.

¹⁸There are inverse floaters that provide a minimum coupon above zero, but we will keep things simple here by assuming the minimum coupon is zero.

Issuer:	U.S. Agency
Principal:	\$100mm
Maturity:	5 years
Coupon:	20% - 4 × 1-year LIBOR
Minimum Coupon:	0.00%
Day Count:	30/360
Resets and Payments:	Annually
Issue Price:	100%

Table 7.6: Terms of an Inverse Floating Rate Note

On a principal amount of \$100mm, this would mean that the coupon payment corresponding to a reset of 5.01% would be $-\$40,000$ without the minimum coupon restriction. Of course, the coupon can't actually be a negative amount, and that is precisely why there is a minimum coupon restriction in the bond. A cap with a notional of \$100mm struck at 5% would pay out $\max(5.01\% - 5\%, 0) \times \text{Notional} = \$10,000$ when LIBOR sets at 5.01%. Since we need the option to pay out \$40,000 in this case, we need four times as much cap, or a notional of \$400mm in order to meet the minimum coupon guarantee. For every bp that LIBOR sets above 5%, the quantity $20\% - 4 \times 1\text{-year LIBOR}$ will decrease by 4 bps, or \$40,000 for a principal amount of \$100mm. Thus, a notional of \$400mm on the cap is required so that for every bp above the strike of 5% the option pays out the necessary amount of \$40,000.

To see this a bit more formally, note that we can write the coupon of the bond as

$$\max(20\% - 4 \times \text{LIBOR}, 0).$$

That is, the bond pays $20\% - 4 \times \text{LIBOR}$, except when this quantity is negative, in which case it pays zero. Notice that¹⁹

$$\begin{aligned} \max(20\% - 4 \times \text{LIBOR}, 0) &= 20\% - 4 \times \text{LIBOR} + \max(4 \times \text{LIBOR} - 20\%, 0) \\ &= 20\% - 4 \times \text{LIBOR} + 4 \times \max(\text{LIBOR} - 5\%, 0). \end{aligned} \quad (7.3)$$

¹⁹This first equality follows from the identity $\max(A, B) = A + \max(0, B - A)$, where we have set $A = 20\% - 4 \times \text{LIBOR}$ and $B = 0$.

Coupon Payment Date (in years)	Assumed Forward Rate	Projected Coupon w/o Min Coupon Restriction	Projected Coupon w/ Min Coupon Restriction
1.0	3.00%	8.00%	8.00%
2.0	4.00%	4.00%	4.00%
3.0	4.50%	2.00%	2.00%
4.0	5.00%	0.00%	0.00%
5.0	5.50%	-2.00%	0.00%

Table 7.7: Assumed Forward Rates and Projected Coupons of Leveraged Inverse Floater in Table 7.6 Both With and Without Minimum Coupon Restriction

The last term in Equation 7.3 is a cap struck at 5% multiplied by a leverage factor of four. This cap ensures that the coupon will be zero in the event that the quantity $20\% - 4 \times \text{LIBOR}$ is negative.

Suppose that we buy the note whose terms are outlined in Table 7.6. Further suppose that today's 1-year LIBOR set is 3%, and suppose that we bootstrap a swap curve (which is not specified here) and obtain forward LIBOR rates as shown in Table 7.7. Based on these assumed forwards, the projected coupons without the minimum coupon restriction of zero and with the minimum coupon restriction are also presented.

As this table demonstrates, when the forwards are steep (which occurs when the spot swap curve is steep, as it was in the early 1990s when inverse floaters were extremely popular), the projected coupons paid by the bond decline quickly. The near-term coupons must compensate for the predicted below market coupons that the investor is expected to obtain in the future. When we have coupons of the form

$$\max(C\% - 4 \times \text{LIBOR}, 0),$$

the level of C is such that it balances out the cash flows that the bondholder is projected to receive over the life of the bond. When the curve is upward sloping, the current coupon income received from an inverse floater will be above market in order to compensate for the below market coupons that the forwards are predicting that the investor will receive later on. In the case of the hypothetical forwards in Table 7.7, the investor is receiving a current coupon of 8% in an environment where short-term rates are only 3%! This is a huge pickup in yield. Inasmuch as the investor believes that the forwards are wrong, and that future LIBOR rates will be lower than implied

Coupon Payment Date (in years)	Assumed Forward Rate	Projected Coupon w/o Min Coupon Restriction	Projected Coupon w/ Min Coupon Restriction
1.0	5.00%	0.00%	0.00%
2.0	5.75%	−3.00%	0.00%
3.0	6.50%	−6.00%	0.00%
4.0	6.75%	−7.00%	0.00%

Table 7.8: Hypothetical Forward Rates a Year After the Leveraged Inverse Floater in Table 7.6 Was Purchased, and Projected Coupons Both With and Without Minimum Coupon Restriction

by the forwards, the bond will provide higher coupons than those projected in Table 7.7.

An investment in an inverse floater in a steep yield curve environment is therefore a play against the realization of forward rates. The investor will receive above-market current coupon income from the bond based on the (relatively) low level of the underlying reference rate and is taking the view that future spot rates of the underlying reference rate are not as high as predicted by the forwards. This is precisely the view that Orange County was expressing, as we will discuss in Section 7.7.2.

If, however, the investor is wrong, and rates rise dramatically, the effect on the value of a leverage inverse floater can be devastating. For example, suppose that a year after buying the leverage inverse floater that pays coupons of $\max(20\% - 4 \times \text{LIBOR}, 0)$, rates spike up dramatically as shown in Table 7.8. In this example, the then-current setting of 1-year LIBOR (a year after the bond has been purchased) is 5%, and the forward curve is very steep, reflecting the market's expectation that rates will rise even further. In this case, the projected coupons on the bond are all zero according to the forwards. The price of the bond will be very close to the price of a zero coupon bond of the same maturity.

7.7.1 An Example: Pricing Up a Leveraged Inverse Floater

Recall now our stylized example in which the swap curve is quoted annual bond versus 1-year LIBOR. Suppose the swap curve is as shown in Table 7.5 and the base vol curve is flat at 15%. Suppose an investor wants to purchase a 4-year leveraged inverse floater with a coupon of the form

$Y\% - 2 \times 1\text{-year LIBOR}$, subject to a minimum coupon of 0.00%. Assume the issuer pays LIBOR flat. What level of Y can be provided to the investor in the note?

In the underlying swap transaction, the swap dealer will be receiving LIBOR flat and will be paying coupons of the form $Y\% - 2 \times 1\text{-year LIBOR}$, subject to the minimum coupon of 0.00%. That is, the dealer will be paying coupons equal to

$$\max(Y\% - 2 \times 1\text{-year LIBOR}, 0).$$

Notice that

$$\begin{aligned} \max(Y\% - 2 \times \text{LIBOR}, 0) \\ &= Y\% - 2 \times \text{LIBOR} + \max(2 \times \text{LIBOR} - Y\%, 0) \\ &= Y\% - 2 \times \text{LIBOR} + 2 \times \max(\text{LIBOR} - Y\%/2, 0). \end{aligned} \quad (7.4)$$

Thus, we can equivalently view this as the dealer paying coupons of the form $Y\% - 2 \times 1\text{-year LIBOR}$ (without the minimum coupon restriction) and selling \$200mm of a LIBOR cap struck at $Y\%/2$. By selling the cap, the dealer will ensure that the coupons the investor receives will never go below 0%. We iterate to find the value of Y such that the value of the cash flows

$$Y\% - 2 \times 1\text{-year LIBOR}$$

plus the value of the cap are equal to the value of the LIBOR flat flows that the dealer receives. As shown in Figure 7.5, we find that $Y = 16.346\%$.

Interest Rate Sensitivity of Leveraged Inverse Floaters

Now that we have structured the bond, let's take a look at how the value of the bond will change given changes in interest rates. In Figure 7.6, we show how the price of the bond varies given an instantaneous parallel shift of the swap curve.²⁰ As expected, as interest rates go down, the value of the bond goes up, and vice versa. It is important to note that the interest rate sensitivity of the bond changes with the level of interest rates.

The metric that is most often employed to examine the price sensitivity of bonds to a change in interest rates is duration. In particular, for bonds

²⁰Once the terms of the bond have been established, we can determine the value of the bond given a parallel shift in the swap curve simply by shifting the swap curve and looking at the value of the cash flows of the bond in the new rate environment. This valuation also considers the value of the embedded cap in the bond. We have made the assumption that implied vol does not change as we shift the curve. We have also made the assumption that the first coupon of the bond varies as we shift the curve.

STRUCTURED NOTES

Time	zc	Forward	Float	PV Float	Y – 2L	CF(Y – 2L)	PV(Y – 2L)
1	5.000%	5.000%	5,000,000	4,761,905	6.346%	(6,346,264)	(6,044,061)
2	5.113%	5.226%	5,225,748	4,729,739	5.895%	(5,894,767)	(5,335,257)
3	5.312%	5.712%	5,712,202	4,890,656	4.922%	(4,921,859)	(4,213,982)
4	5.494%	6.041%	6,041,364	4,877,792	4.264%	(4,263,535)	(3,442,374)
				19,260,092			(19,035,674)
						\$200mm cap	(224,418)
		Y	16.346%			NPV	0
		Notional	100,000,000				

Cap model									
Time	zc	Period	Forward	σ	d_1	d_2	Forward premium	Caplet	Caplet in dollars
1	5.000%	0x1	5.000%	n/a	n/a	n/a	0.000%	0.000%	0
2	5.113%	1x2	5.226%	15%	-2.907	-3.057	0.000%	0.000%	361
3	5.312%	2x3	5.712%	15%	-1.583	-1.795	0.027%	0.023%	23,224
4	5.494%	3x4	6.041%	15%	-1.033	-1.293	0.110%	0.089%	88,625
Strike 8.173%			Notional		100,000,000			112,209	
									in bps 11.2

Figure 7.5: Computing the Coupon of the Leveraged Inverse Floater

with embedded options (such as a leveraged inverse floater), we typically use *effective duration*.²¹ The effective duration of the bond is given by

$$\frac{P_{-\Delta y} - P_{+\Delta y}}{2 \times \Delta y \times P_0}, \quad (7.5)$$

where $P_{-\Delta y}$ and $P_{+\Delta y}$ are the value of the bond when the swap curve is shifted down or up by an amount Δy , respectively, and P_0 represents the

²¹For a review of effective duration, see Appendix B.